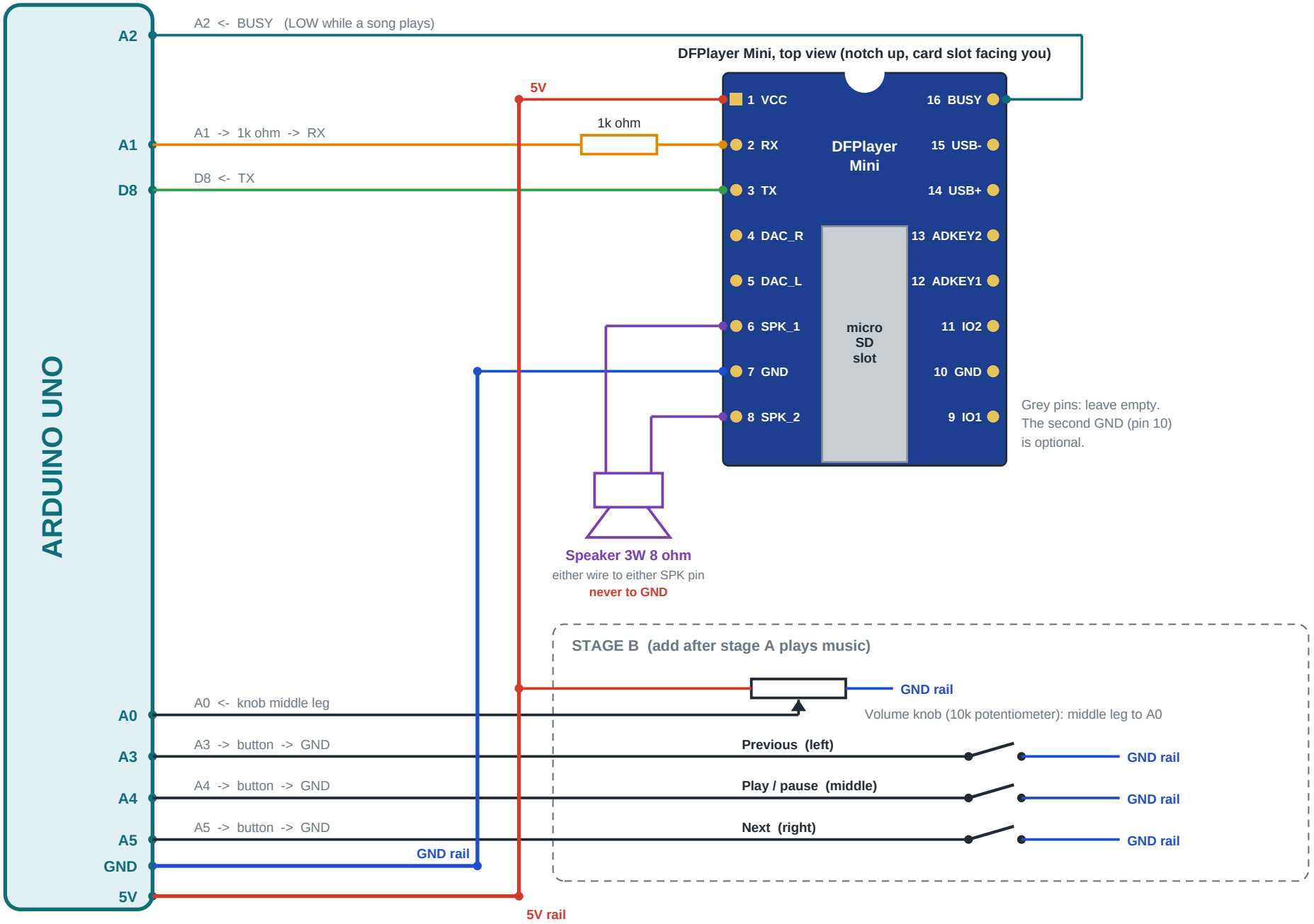
RFID Music Box - Phase 2 Wiring

Arduino Uno | DFPlayer Mini + speaker (stage A), volume knob + 3 buttons (stage B) | Sketch: src/main.cpp

Phase 1 wiring (RC522 reader on D9-D13, LCD on D2-D7) stays exactly as in phase-1-rfid-lcd.pdf. This page only shows what phase 2 adds.



Lines that cross without a dot are not connected.

DFPlayer Mini - Pin Guide

How to find each pin on the real module. Source: DFRobot DFR0299 wiki and the FN-M16P datasheet (same pinout).

card slides in here, from the bottom edge

How to hold it

1. Turn it so the **microSD slot faces you** (you see the metal card cage).
2. Turn it so the **small half-moon notch is at the top** and the slot's opening is at the bottom.
3. Now **pin 1 (VCC) is the top-left pin**. The left side counts 1-8 going down; the right side counts 16-9 going down, so **BUSY (16) is top-right**.
4. Check: flip it over. Pin 1 has a **square** solder pad; all the others are round.

Every pin

Pin	Name	What it does	In this project
1	**VCC**	Power in, 3.2-5V	5V rail
2	**RX**	Serial in (3.3V level)	Uno A1 through 1k ohm
3	**TX**	Serial out (3.3V level)	Uno D8
4	**DAC_R**	Line out, right (headphones / amp)	leave empty
5	**DAC_L**	Line out, left (headphones / amp)	leave empty
6	**SPK_1**	Speaker + (built-in 3W amp)	Speaker wire
7	**GND**	Ground	GND rail
8	**SPK_2**	Speaker - (built-in 3W amp)	Speaker wire
9	**IO1**	Button: short = previous, long = volume down	leave empty
10	**GND**	Ground	GND rail
11	**IO2**	Button: short = next, long = volume up	leave empty
12	**ADKEY1**	Resistor-ladder buttons	leave empty
13	**ADKEY2**	Resistor-ladder buttons	leave empty
14	**USB+**	USB data + (flash drive)	leave empty
15	**USB-**	USB data - (flash drive)	leave empty
16	**BUSY**	LOW while playing, HIGH when stopped	Uno A2

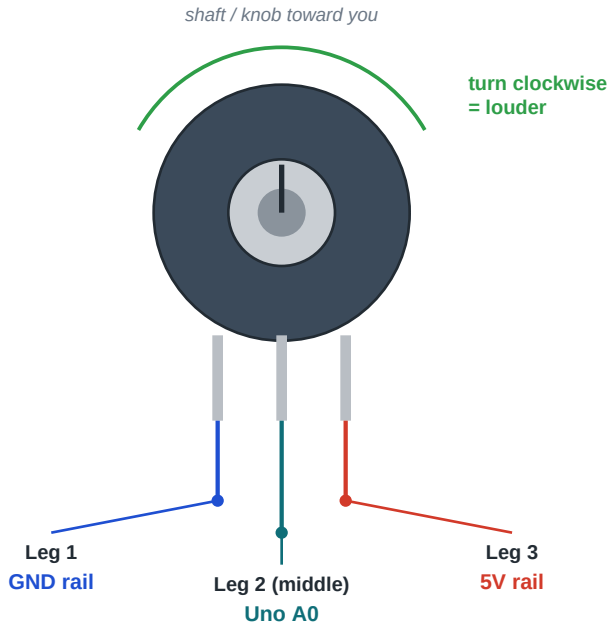
Specs that matter here

Power: 3.2-5V, so the Uno's 5V is fine. **Serial:** 9600 baud, 3.3V logic: the 1k resistor on RX protects it from the Uno's 5V and cuts hiss. **Speaker:** under 3W, 4-8 ohm, straight onto SPK_1/SPK_2. **Card:** microSD up to 32GB, FAT16/FAT32. **Files:** MP3 or WAV, 8-48kHz; folders **01-99**, songs **001.mp3-255.mp3** (a name may continue after the number, e.g. 001Dino.mp3).

Stage B - Volume Knob and Buttons

How the potentiometer and push buttons work inside, and which leg goes where.

Potentiometer (volume knob)



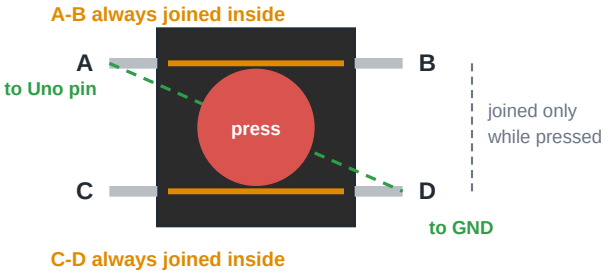
Inside, a strip of resistance runs between **legs 1 and 3**, and the **middle leg** slides along it as you turn. With the outer legs on GND and 5V, the middle leg gives the Uno anything from 0V to 5V. **A0** reads that as 0 to 1023, and the sketch turns it into volume 0 to MAX_VOLUME.

Louder goes the wrong way? Swap the two outer legs. The middle leg always stays on A0.

On the breadboard: push the three legs into three different rows, then run one jumper from each row.

Schematic symbol

Push button (x3)

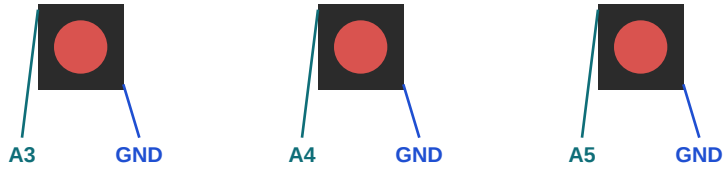


The button has 4 legs but only 2 connections: **A and B are always joined**, and so are **C and D**. Pressing joins the two pairs. Which pair is which depends on the button, so use the safe trick:

Use two diagonal legs (A and D, or B and C). Diagonal legs are always on different pairs, so it works whichever way the button is turned. One diagonal leg goes to the Uno pin, the other to GND.

On the breadboard: put each button across the centre gap. No resistor is needed: the sketch turns on the Uno's built-in pull-up, so the pin reads HIGH normally and LOW while pressed.

Left to right on the box



Schematic symbol

RFID Music Box - Phase 2 Parts & Connections

Everything on page 1 as lists, plus a few tips.

Parts

Qty	Part	Notes
1	DFPlayer Mini (DFRobot)	Straddle the breadboard's centre gap, card slot facing up.
1	microSD card, 32GB or less	MS-DOS (FAT). Folders 01, 02, ... with 001.mp3, 002.mp3, ...
1	Speaker, 3W 8 ohm	Cut off the JST plug or use a JST-PH socket.
1	1k ohm resistor	Brown, black, red. Between A1 and RX.
1	10k potentiometer	Stage B. Volume knob.
3	Push buttons	Stage B. Previous, play/pause, next.
~14	Jumper wires	

Connections

Uno pin	Goes to	Notes
5V rail	DFPlayer VCC (pin 1)	Top-left pin.
GND rail	DFPlayer GND (pin 7)	
D8	DFPlayer TX (pin 3)	Uno listens here.
A1	1k ohm, then DFPlayer RX (pin 2)	Uno talks here.
A2	DFPlayer BUSY (pin 16)	Top-right pin.
-	Speaker on SPK_1 (6) and SPK_2 (8)	Either way round.
A0	Knob middle leg	Stage B. Outer legs: 5V, GND.
A3 / A4 / A5	Previous / play-pause / next button	Stage B. Other leg to GND.

If something is wrong

LCD says MP3 not found: the Uno got no answer from the DFPlayer. Most often TX and RX are swapped (DFPlayer TX goes to D8). Also check the SD card is pushed in and formatted MS-DOS (FAT), then press reset.

Card shows its title but no sound: check the speaker is on SPK_1/SPK_2 (pins 6 and 8), not on DAC_L/DAC_R.

Only the first song plays: the BUSY wire (pin 16 to A2) is missing or on the wrong pin.

Songs play in the wrong order: files must be named 001.mp3, 002.mp3 inside 01, 02 ... Run dot_clean on the card after copying.

Hiss or crackle: make sure the 1k resistor is in the RX line and every GND goes to the same rail.

Card shows New card: its ID isn't in src/cards.h yet. Copy the ID into a new line and upload.

What it does

Tap a card from cards.h: its folder plays from song 1, the LCD shows the title and **Song 1**, and the next song starts on its own. After the last song it shows **Tap a card...** again. Another card switches folders; the same card again does nothing. An unknown card shows **New card:** and its ID for 3 seconds while the music keeps playing. Stage B: the knob sets volume 0-30; **Previous** / **Play-pause** / **Next** buttons; paused shows **Paused 2**. Next on the last song goes back to song 1.